

Motion-Aware First-Boundary FoV Estimation

Revised roadmap for the Probabilistic Robotics semester project

1. Goal and scope

Goal. Recursively estimate the first-occlusion boundary from noisy, discrete range rays while compensating for observer translation and rotation. The output is a piecewise boundary and confidence information that can be supplied to the existing FoV-maintenance controller. Developing or proving the controller is outside this project.

Scope. Work in 2D with static opaque polygonal obstacles, known observer motion, and simulated first-return range sensing co-located with the FoV. The simulator and evaluator know the polygons; the estimator receives only observer motion and range returns. There is no occupancy grid, SLAM, persistent environment map, hidden-surface model, or scan archive.

2. Boundary state and Bayes-filter notation

For each sensor-relative ray angle θ , let $B_t(\theta)$ denote the distance to the *first* occluding surface at time t . A point at polar coordinates (ρ, θ) is visible when $\rho \leq B_t(\theta)$, subject to the nominal FoV range and aperture.

Choose a fixed angular output grid $\Theta = \{\theta_1, \dots, \theta_N\}$ and define the boundary state

$$\mathbf{b}_t = \begin{bmatrix} B_t(\theta_1) & \cdots & B_t(\theta_N) \end{bmatrix}^T. \quad (1)$$

In the notation of Thrun, Burgard, and Fox [1], the posterior belief is

$$\text{bel}(\mathbf{b}_t) = p(\mathbf{b}_t \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}), \quad (2)$$

where $\mathbf{u}_t = [v_x, v_y, \omega]^T$ is known observer motion and $\mathbf{z}_t$ contains the valid first-return ranges. The prediction and correction steps are

$$\overline{\text{bel}}(\mathbf{b}_t) = \int p(\mathbf{b}_t \mid \mathbf{u}_t, \mathbf{b}_{t-1}) \text{bel}(\mathbf{b}_{t-1}) d\mathbf{b}_{t-1}, \quad (3)$$

$$\text{bel}(\mathbf{b}_t) = \eta p(\mathbf{z}_t \mid \mathbf{b}_t) \overline{\text{bel}}(\mathbf{b}_t). \quad (4)$$

Only this current observer-frame belief, its uncertainty, and surface-support labels are retained. When a surface leaves the FoV or becomes occluded, it is forgotten rather than stored for later re-use.

3. Motion model for supported boundary pieces

Let v_x, v_y be translational velocity expressed in the observer frame and let ω be counterclockwise angular velocity. For a fixed physical point on a visible surface with polar coordinates (r, θ) in the observer frame,

$$\dot{r} = -v_x \cos \theta - v_y \sin \theta, \quad \dot{\theta} = \frac{v_x \sin \theta - v_y \cos \theta}{r} - \omega. \quad (5)$$

On a smooth surface piece that remains the first surface, $r(t) = B(\theta(t), t)$. The chain rule yields the local boundary transport equation

$$\frac{\partial B}{\partial t} = -v_x \cos \theta - v_y \sin \theta + \left(\omega - \frac{v_x \sin \theta}{B} + \frac{v_y \cos \theta}{B} \right) \frac{\partial B}{\partial \theta}. \quad (6)$$

For pure rotation, $\partial B / \partial t = \omega \partial B / \partial \theta$: the range profile shifts along the angular axis. Equation (6) is related to range-flow geometry [2]; this project uses known motion to predict range, rather than estimating motion from consecutive scans.

4. Practical prediction and EKF approximation

Start with a geometric version of the prediction before implementing its covariance. For each supported boundary sample, form its point in the old observer frame,

$$\mathbf{s}_{t-1,i} = \hat{B}_{t-1}(\theta_i) \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix}. \quad (7)$$

For known displacement $\Delta \mathbf{p}$, expressed in that frame, and heading change $\Delta \psi$,

$$\mathbf{s}_{t,i}^- = \mathcal{R}(-\Delta \psi)(\mathbf{s}_{t-1,i} - \Delta \mathbf{p}). \quad (8)$$

Recover its new bearing and range, then resample only within the same supported surface segment. Never interpolate through a detected depth jump. The result defines a nonlinear transition function

$$\mathbf{b}_t = f(\mathbf{b}_{t-1}, \mathbf{u}_t) + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t). \quad (9)$$

Approximate the belief as Gaussian:

$$\text{bel}(\mathbf{b}_t) \approx \mathcal{N}(\hat{\mathbf{b}}_t, P_t). \quad (10)$$

The EKF prediction is

$$\hat{\mathbf{b}}_t^- = f(\hat{\mathbf{b}}_{t-1}, \mathbf{u}_t), \quad (11)$$

$$P_t^- = F_t P_{t-1} F_t^T + Q_t, \quad F_t = \left. \frac{\partial f}{\partial \mathbf{b}} \right|_{\mathbf{b}_{t-1}, \mathbf{u}_t}. \quad (12)$$

Use local Jacobians only while the segment and interpolation associations are unchanged. Q_t represents finite-resolution transport, interpolation error, and unmodeled exposure events. If the motion estimate is uncertain, add $G_t \Sigma_{u,t} G_t^T$, with $G_t = \partial f / \partial \mathbf{u}$.

5. Range measurement update and reset policy

For valid first returns, use the simple range observation model

$$\mathbf{z}_t = H_t \mathbf{b}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, R_t). \quad (13)$$

Here H_t selects measured angular bins or linearly interpolates within a segment. The correction is linear even though the transition is nonlinear:

$$K_t = P_t^- H_t^T (H_t P_t^- H_t^T + R_t)^{-1}, \quad (14)$$

$$\hat{\mathbf{b}}_t = \hat{\mathbf{b}}_t^- + K_t (\mathbf{z}_t - H_t \hat{\mathbf{b}}_t^-), \quad (15)$$

$$P_t = (I - K_t H_t) P_t^- (I - K_t H_t)^T + K_t R_t K_t^T. \quad (16)$$

A newly exposed surface cannot be predicted from the old first boundary. Mark it unsupported, remove stale cross-correlations, and give its corresponding covariance block a broad but finite prior. For a directly measured scalar bin,

$$K_i = \frac{P_i^-}{P_i^- + R_i}. \quad (17)$$

Thus $P_i^- \gg R_i$ implies $K_i \approx 1$, so the posterior follows the new range return. With no useful prior support, initialize directly: $\hat{B}_i = z_i$, $P_i = R_i$. A high covariance in an *unmeasured* direction creates no information; it must remain unsupported until an observation or a justified within-segment interpolation constrains it.

No-return rays. Keep a separate maximum-range/support flag. A no-return is censored information, not an obstacle exactly at maximum range; initially it should not be assimilated as a Gaussian equality.

6. Sparse rays and interface to the controller

The number N of output samples may exceed the number M of physical rays. History can transport an observed surface to a new bearing, while the new rays correct only observed directions. Start with piecewise-linear interpolation and local covariance coupling *within* segments. Do not globally smooth $B(\theta)$: the true first boundary generally has jumps at obstacle silhouettes.

Return the estimate $\hat{B}_t(\theta)$, standard deviation $\sigma_t(\theta)$, support flags, and segment boundaries. On supported Gaussian portions, the pointwise lower confidence boundary is

$$B_{\epsilon,t}(\theta) = \hat{B}_t(\theta) - \Phi^{-1}(1 - \epsilon)\sigma_t(\theta). \quad (18)$$

Intersect this with the nominal FoV. Unsupported portions should not be asserted visible. This is a pointwise, model-dependent confidence boundary—not a simultaneous guarantee over the whole FoV or trajectory.

The existing controller can consume the resulting segmented boundary or visibility polygon. Confirm whether its distance routine only needs the set itself (preferred) or derivatives of $B(\theta)$; the latter require treating smooth pieces and depth jumps separately. No controller redesign is required for this roadmap.

7. Implementation order and evaluation

1. Generate dense ray-cast ground truth and noisy sparse scans. Validate fixed-view filtering, segmentation, and no-return handling.
2. Validate deterministic rotation and translation transport on surfaces that remain visible. Add EKF covariance propagation and the support/reset policy.
3. Reduce ray count and assess history-assisted reconstruction. Supply the final first boundary to the existing controller for a demonstration.

Compare: (i) current-scan interpolation, (ii) temporal filtering without motion compensation, and (iii) the motion-aware EKF. Use scenarios with rotation, translation toward and along walls, passing obstacle corners, FoV exit/re-entry, and thin occluders. Sweep ray count, range noise, speed, and confidence level.

Report: finite-return boundary RMSE, confidence coverage, false-visible rate, supported angular fraction, error/recovery time after an exposure event, and runtime. Report unsupported directions explicitly so a method cannot improve its apparent error merely by withholding predictions.

8. Literature position

The exact combination proposed here was not found in a targeted literature check: current first-boundary belief; observer-motion EKF prediction; explicit forgetting/resetting; sparse-ray correction; and downstream FoV maintenance. This is a *possible* gap, not a confirmed novelty claim.

The constituent ideas have precedents. Range-flow geometry relates scan change to sensor motion [2]. Zhou et al. [3] develop occlusion-aware visibility CBF control, but their FoV is ray-traced from a SLAM occupancy map and their EKF tracks the evader. Gao et al. [4] use a known obstacle map and an EKF variant for uncertain target visibility. GP-Frontier [5] uses an uncertain local angular range representation for mapless navigation. None of these sources formulates the current first boundary itself as the EKF state in the map-free FoV-estimation pipeline proposed here.

References

- [1] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005. Bayes-filter notation and recursive estimation background.
- [2] M. Jaimez, J. G. Monroy, and J. Gonzalez-Jimenez, “Planar Odometry from a Radial Laser Scanner: A Range Flow-based Approach,” in *ICRA*, 2016. https://github.com/MAPIRlab/rf2o_laser_odometry
- [3] M. Zhou et al., “Control Strategies for Pursuit-Evasion Under Occlusion Using Visibility and Safety Barrier Functions,” in *ICRA*, 2025. <https://arxiv.org/html/2411.01321v2>
- [4] H. Gao et al., “Probabilistic Visibility-Aware Trajectory Planning for Target Tracking in Cluttered Environments,” in *ACC*, 2024. <https://arxiv.org/html/2306.06363v2>
- [5] M. Ali and L. Liu, “GP-Frontier for Local Mapless Navigation,” in *ICRA*, 2023. <https://arxiv.org/pdf/2307.11717>